

Section A — MCQs

1. **C — Dipole-Dipole Forces**
HCl is a polar molecule, so polar molecules attract each other through dipole-dipole forces.
2. **D — Limiting reactant**
The reactant that finishes first decides the maximum yield.
3. **C — 2 moles**
Given: 32 g O₂ = 1 mole O₂.
Reaction: 2H₂ + O₂ → 2H₂O
So 1 mole O₂ produces **2 moles H₂O**.
4. **B — High Surface Tension**
Water has unusually high surface tension due to strong hydrogen bonding.
5. **B — Heat of combustion**
Complete burning of 1 mole of a substance in oxygen gives heat of combustion.
6. **D — O₂ (O₃ is the typo here)**
Elements in their standard state have zero standard enthalpy of formation.
7. **D — Lowering activation energy**
Catalyst provides an alternate path with lower activation energy.
8. **A — 1**
Large jump from 495 to 4560 shows only 1 valence electron.
9. **C — One proton**
Conjugate acid-base pairs differ by one H⁺.
10. **C — Group number**
For representative elements, group number helps determine valence electrons.
11. **B — Propane**
Propane is an alkane and has no functional group.
12. **B — Free radical substitution**
Methane reacts with chlorine in sunlight by free radical substitution.
13. **C — Inductive effect of alkyl groups**
Alkyl groups stabilize carbocations by +I effect.
14. **C — Tollens' reagent**
Tollens' reagent distinguishes aldehydes from ketones.
15. **A — Carboxylic acids**
Aldehydes oxidize to carboxylic acids.
16. **A — f**
f-subshell can hold maximum 14 electrons.
17. **A — I₂**
Iodine is solid at room temperature.

Answer Key:

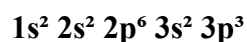
1-C, 2-D, 3-C, 4-D, 5-B, 6-D, 7-D, 8-A, 9-C, 10-C, 11-B, 12-B, 13-C, 14-C, 15-A, 16-A, 17-A.

Section B — Short Questions

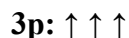
Q2(i) Electronic configuration of P and C according to Hund's rule

Phosphorus, P = 15

Electronic configuration:



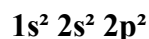
Orbital form:



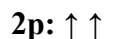
Reason: According to Hund's rule, electrons first occupy degenerate orbitals singly with parallel spin.

Carbon, C = 6

Electronic configuration:



Orbital form:



Reason: The two p-electrons remain unpaired in separate p-orbitals before pairing.

OR: Three differences between sigma and pi bonds

Sigma bond	Pi bond
Formed by head-on overlap	Formed by sidewise overlap
Stronger bond	Weaker bond
Allows free rotation	Restricts free rotation

Example: In ethene, C=C has **one sigma bond and one pi bond**.

Q2(ii) How azimuthal quantum number determines electrons in d-subshell?

Azimuthal quantum number is represented by **l**.

For d-subshell:

$$l = 2$$

Number of orbitals:

$$2l + 1 = 2(2) + 1 = 5$$

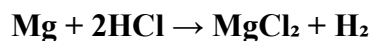
Each orbital holds 2 electrons, so:

$$5 \times 2 = 10 \text{ electrons}$$

Therefore, d-subshell can accommodate **maximum 10 electrons**.

OR: Mass of Mg needed to produce 100 cm³ H₂ at STP

Reaction:



At STP:

$$1 \text{ mole gas} = 22,400 \text{ cm}^3$$

Moles of H₂:

$$100 / 22400 = 0.00446 \text{ mol}$$

From equation:

1 mol Mg produces 1 mol H₂

So moles of Mg = 0.00446 mol

Mass of Mg:

$$\text{mass} = \text{moles} \times \text{molar mass}$$

$$= 0.00446 \times 24 = 0.107 \text{ g}$$

Answer: 0.107 g Mg

Q2(iii) Why bond energy of F—F is lower than Cl—Cl although F size is smaller?

F atom is very small, so in F₂ the lone pairs of both fluorine atoms come very close.

This causes strong **lone pair-lone pair repulsion**, which weakens the F—F bond.

Therefore, F—F bond energy is less than Cl—Cl bond energy.

OR: Why heat of vaporization ΔH_v is always greater than heat of fusion ΔH_f ?

In fusion, solid changes into liquid. Only some intermolecular forces are weakened.

In vaporization, liquid changes into gas. Almost all intermolecular forces are overcome.

Therefore, more energy is required for vaporization.

So:

$$\Delta H_v > \Delta H_f$$

Q2(iv) Protons, neutrons and electrons in Na⁺ and Cl⁻

For sodium:

Atomic number = 11

Mass number = 23

Na⁺ has lost one electron.

Protons = 11

Electrons = 10

Neutrons = 23 - 11 = 12

For chlorine:

Atomic number = 17

Mass number = 35.5 is average, but commonly Cl-35 is used.

Cl⁻ has gained one electron.

Protons = 17
Electrons = 18
Neutrons = $35 - 17 = 18$

OR: Three uses of sulphur in industry/daily life

1. Used in the manufacture of **sulphuric acid**.
 2. Used in **vulcanization of rubber**.
 3. Used in **fungicides and medicines**.
-

Q2(v) Type of bond in O₂, KCl, HCl and H₂O

O₂: Covalent non-polar bond
Reason: Same atoms share electrons equally.

KCl: Ionic bond
Reason: K transfers electron to Cl.

HCl: Polar covalent bond
Reason: Shared electrons are attracted more by Cl.

H₂O: Polar covalent bond
Reason: O is more electronegative than H.

OR: K_{sp} expressions

For **CaF₂**:



$$\text{K}_{\text{sp}} = [\text{Ca}^{2+}][\text{F}^-]^2$$

For **MgSO₄**:



$$\text{K}_{\text{sp}} = [\text{Mg}^{2+}][\text{SO}_4^{2-}]$$

Q2(vi) Limiting reactant: 10 g A reacts with 20 g B

Reaction:



Given:

Moles of A = $10 \text{ g} / 27 = 0.370 \text{ mol}$

Moles of B = $20 \text{ g} / 19 = 1.053 \text{ mol}$

According to reaction:

2 mol A need 3 mol B

For 0.370 mol A, B required:

$$= 0.370 \times 3/2 = 0.555 \text{ mol B}$$

Available B = 1.053 mol, which is more than required.

So A is limiting reactant.

Product AB formed:

2 mol A produce 2 mol AB

So moles of AB = 0.370 mol

Molar mass of AB = $27 + 19 = 46 \text{ g/mol}$

Mass of AB:

$$= 0.370 \times 46 = 17.02 \text{ g}$$

Answer: A is limiting reactant and 17.02 g AB is formed.

OR: Justify

Atomic size decreases along the period

Across a period, nuclear charge increases but shielding remains almost same. So electrons are pulled more strongly towards nucleus.

Therefore, atomic size decreases.

Metallic character increases along the group

Down the group, atomic size increases and ionization energy decreases. Atoms lose electrons more easily.

Therefore, metallic character increases.

Q2(vii) Crystalline and amorphous solids

Crystalline solids:

Particles are arranged in regular pattern. They have sharp melting point.

Examples: NaCl, diamond, sugar.

Amorphous solids:

Particles are irregularly arranged. They do not have sharp melting point.

Examples: glass, rubber, plastic.

OR: General formulas of primary, secondary and tertiary amines

Primary amine: $R-NH_2$

Example: CH_3NH_2

Secondary amine: $R-NH-R$

Example: CH_3NHCH_3

Tertiary amine: R_3N

Example: $N(CH_3)_3$

Q2(viii) Exothermic and endothermic reactions

Exothermic reaction:

Reaction in which heat is released.

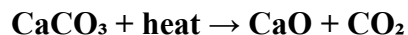
Example:



Endothermic reaction:

Reaction in which heat is absorbed.

Example:

**OR: Catenation**

Catenation is the ability of an element to form covalent bonds with itself, making chains or rings.

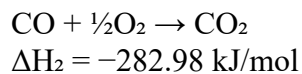
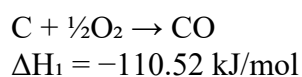
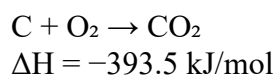
Carbon shows strong catenation because:

1. C—C bond is strong.
2. Carbon has small atomic size.
3. Carbon has valency 4.

Importance: It helps carbon form millions of organic compounds.

Q2(ix) Hess's law energy cycle

Given:



According to Hess's law:

$$\Delta H = \Delta H_1 + \Delta H_2$$

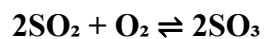
$$= -110.52 + (-282.98)$$

$$= \mathbf{-393.50 \text{ kJ/mol}}$$

So the two-step path gives the same enthalpy change as direct conversion.

OR: K_p calculation

Reaction:



Given:

$$[\text{SO}_2] = 0.59 \text{ M}$$

$$[\text{O}_2] = 0.05 \text{ M}$$

$$[\text{SO}_3] = 0.259 \text{ M}$$

Expression:

$$K_c = [\text{SO}_3]^2 / [\text{SO}_2]^2 [\text{O}_2]$$

$$\Delta n = \text{moles products} - \text{moles reactants} \\ = 2 - 3 = -1$$

Formula:

$$K_p = K_c(RT)^{\Delta n}$$

First calculate K_c:

$$K_c = (0.259)^2 / (0.59)^2(0.05)$$

$$= 0.067081 / 0.017405$$

$$= \mathbf{3.85}$$

$$\text{Temperature} = 450^\circ\text{C} = 723 \text{ K}$$

$$K_p = K_c(RT)^{-1}$$

$$= 3.85 / (0.0821 \times 723)$$

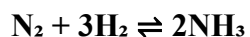
$$= 3.85 / 59.36$$

$$= \mathbf{0.0649}$$

Answer: K_p ≈ 0.065

Q2(x) Effect of adding NH₃ in equilibrium

Reaction:



a. Ammonia removed from mixture

Equilibrium shifts forward to produce more NH₃.

b. More nitrogen added

Equilibrium shifts forward because reactant concentration increases.

OR: Preparation of alkyl halides

From alkenes

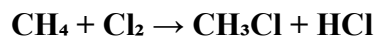
Example:



Ethene reacts with HBr to form ethyl bromide.

From alkanes

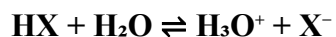
Example:



Condition: sunlight/UV light.

Q2(xi) Acid dissociation constant K_a

Given:



K_a expression:

$$\text{K}_a = [\text{H}_3\text{O}^+][\text{X}^-] / [\text{HX}]$$

Water is not included because it is present in large excess and its concentration remains almost constant.

OR: Lucas test

Lucas reagent is:

Conc. HCl + anhydrous ZnCl₂

It distinguishes alcohols by turbidity.

Tertiary alcohol: immediate turbidity

Secondary alcohol: turbidity after few minutes

Primary alcohol: no turbidity at room temperature

Reason: Tertiary alcohol forms stable carbocation quickly.

Q2(xii) Factors affecting reactivity of alkyl halides

1. **Nature of carbon-halogen bond:**

Weaker bond means more reactive.

Order: RI > RBr > RCl > RF

2. **Structure of alkyl group:**

Tertiary alkyl halides are more reactive in SN1 reactions because they form stable carbocations.

OR: 2,4-DNP test

2,4-dinitrophenylhydrazine is used to detect carbonyl compounds.

Aldehydes and ketones give yellow/orange precipitate with 2,4-DNP.

General reaction:

Aldehyde/Ketone + 2,4-DNP → 2,4-dinitrophenylhydrazone + water

It confirms the presence of **C=O group**.

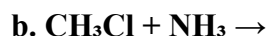
Q2(xiii) Boiling vs evaporation

Boiling	Evaporation
Occurs at fixed temperature	Occurs at all temperatures
Occurs throughout liquid	Occurs only at surface
Fast process	Slow process

OR: Complete reactions



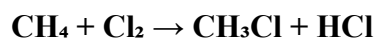
Major product: **propanenitrile / ethyl cyanide**



Major product: **methylamine**

Q2(xiv) Free radical substitution of methane with chlorine

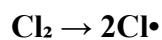
Overall reaction:



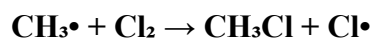
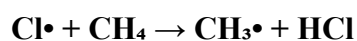
Condition: sunlight/UV light.

Step 1: Initiation

Cl_2 breaks into chlorine free radicals:



Step 2: Propagation



The chlorine radical is regenerated.

Step 3: Termination

Free radicals combine:



OR: Iodoform test

Methanol and ethanol

Ethanol gives positive iodoform test because it contains **CH₃CH(OH)— group** after oxidation.

Reaction observation: yellow precipitate of iodoform, CHI₃.

Methanol does not give this test.

So:

Ethanol: yellow ppt

Methanol: no yellow ppt

Formaldehyde and acetaldehyde

Acetaldehyde gives positive iodoform test because it has **CH₃CHO group**.

Formaldehyde does not.

So:

Acetaldehyde: yellow ppt of CHI₃

Formaldehyde: no ppt

Section C — Long Questions

Q3

What is hybridization? Explain CH_4 and BF_3

Hybridization is the mixing of atomic orbitals of almost equal energy to form new equivalent orbitals called hybrid orbitals.

Methane, CH_4

Carbon ground state:

$1s^2 2s^2 2p^2$

Excited state:

$1s^2 2s^1 2p^3$

One 2s and three 2p orbitals mix to form **four sp^3 hybrid orbitals**.

Each sp^3 orbital overlaps with 1s orbital of hydrogen to form four C—H sigma bonds.

Shape: **tetrahedral**

Bond angle: **109.5°**

Boron trifluoride, BF_3

Boron atomic number = 5

Electronic configuration:

$1s^2 2s^2 2p^1$

In BF_3 , one 2s and two 2p orbitals of boron hybridize to form **three sp^2 hybrid orbitals**.

These three orbitals form three B—F sigma bonds.

Shape: **trigonal planar**

Bond angle: **120°**

OR: Markovnikov's rule and propene with $\text{HBr}/\text{H}_2\text{O}$

Markovnikov's rule:

When an unsymmetrical reagent adds to an unsymmetrical alkene, the positive part of the reagent goes to the carbon already having more hydrogen atoms.

Reaction of propene with HBr

Propene:



HBr adds as H^+ and Br^- .

H^+ attaches to terminal carbon because it already has more hydrogen atoms. This forms a more stable secondary carbocation.

Reaction:



Major product:

2-bromopropane

Minor product:

1-bromopropane

Reason: Secondary carbocation is more stable than primary carbocation.

Reaction of propene with H_2O

In acidic medium:



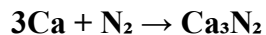
Major product:

propan-2-ol / isopropyl alcohol

Q4

Calculate mass of calcium nitride and percentage yield

Reaction:



Given:

Mass of Ca = 54.9 g

Atomic mass of Ca = 40

Atomic mass of N = 14

Moles of Ca:

$$54.9 / 40 = 1.3725 \text{ mol}$$

From equation:

3 mol Ca produce 1 mol Ca₃N₂

Moles of Ca₃N₂:

$$1.3725 / 3 = 0.4575 \text{ mol}$$

Molar mass of Ca₃N₂:

$$3(40) + 2(14) = 120 + 28 = 148 \text{ g/mol}$$

Mass of Ca₃N₂:

$$0.4575 \times 148 = 67.71 \text{ g}$$

So theoretical yield:

$$67.71 \text{ g}$$

Now actual yield = 0.13 moles Ca₃N₂

Actual mass:

$$0.13 \times 148 = 19.24 \text{ g}$$

Percentage yield:

$$\text{Actual yield} / \text{Theoretical yield} \times 100$$

$$= 19.24 / 67.71 \times 100$$

= 28.41%

Answer:

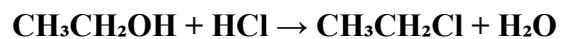
Mass of Ca_3N_2 = 67.71 g

Percentage yield = 28.41%

OR: Preparation of alkyl halides

a. From alcohols using HCl/ZnCl_2

Example:



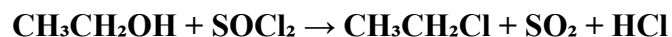
ZnCl_2 acts as a catalyst. This reaction converts alcohol into alkyl chloride.

Major product:

chloroethane

b. From alcohols using SOCl_2

Example:



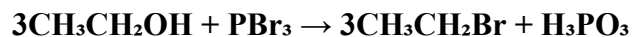
This is a good method because gaseous by-products escape, giving pure alkyl chloride.

Major product:

chloroethane

c. From alcohols using PBr_3

Example:



Major product:

bromoethane

Q5

Maxwell-Boltzmann distribution and effect of temperature

Maxwell-Boltzmann distribution curve shows the distribution of molecular energies in a gas.

At low temperature, fewer molecules have energy equal to or greater than activation energy.

When temperature increases:

1. The curve becomes broader.
2. Peak of curve becomes lower.
3. More molecules have energy greater than activation energy.
4. Number of effective collisions increases.
5. Rate of reaction increases.

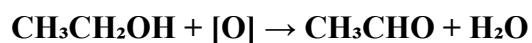
Reason: Reaction occurs only when reacting particles collide with energy equal to or greater than activation energy.

So, increasing temperature increases the fraction of molecules crossing activation energy barrier.

OR: Interconversion reactions

a. Ethanol into acetaldehyde

Oxidation/dehydrogenation:



Product: **ethanal / acetaldehyde**

b. Ethanol into ethyl acetate

Esterification:



Condition: concentrated H_2SO_4

Product: **ethyl acetate / ethyl ethanoate**

c. Ethanol into ethene

Dehydration:



Condition: concentrated H_2SO_4 at about 170°C .

Product: **ethene**

Q6

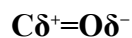
Nucleophilic addition reaction

A **nucleophilic addition reaction** is a reaction in which a nucleophile attacks an electron-deficient carbon atom of a carbonyl group and adds across the $\text{C}=\text{O}$ bond.

Aldehydes and ketones show nucleophilic addition because the carbonyl carbon is partially positive.

$\text{C}=\text{O}$ bond polarity:

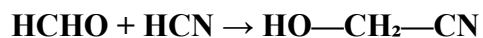
Oxygen is more electronegative, so:



The nucleophile attacks carbonyl carbon.

Mechanism of HCN addition to methanal

Reaction:



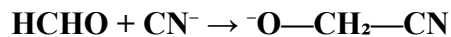
Product:

cyanohydrin

Step 1: HCN produces CN^-



Step 2: CN^- attacks carbonyl carbon



Step 3: Oxygen gets proton



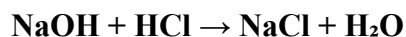
Final product:

2-hydroxyethanenitrile

NaOH as a base

NaOH provides OH^- ions and accepts H^+ ions.

Example:



So NaOH neutralizes acids and behaves as a strong base.

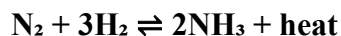
OR: Le Chatelier's principle

Le Chatelier's principle:

If a system at equilibrium is disturbed by changing concentration, pressure or temperature, the equilibrium shifts in a direction that opposes the change.

For exothermic reaction

Example:



To obtain maximum product:

1. Use **low temperature** because heat is a product.
2. Use **high pressure** if product side has fewer gas moles.
3. Remove product continuously.
4. Increase concentration of reactants.

For endothermic reaction

Example:



To obtain maximum product:

1. Use **high temperature** because heat is a reactant.
2. Increase concentration of reactants.
3. Remove product continuously.
4. Pressure effect depends on number of gas moles on both sides.